

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color skyblue for the bars.
- Title the plot as "Passenger Class Distribution".
- Label the x-axis as "`Pclass`" and the y-axis as "Count".

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use lightblue for males and lightcoral for females.
- Include percentages on the slices (use `autopct = '%1.1f%%'`).
- Title the plot as "Gender Distribution".

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use lightgreen for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to 8 for the histogram.
- Title the plot as "Age Distribution".
- Label the x-axis as "`Age`" and the y-axis as "Frequency".

Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.
- Use the colors lightblue for survivors (1) and lightcoral for non-survivors (0).
- Title the plot as "Survival Count".
- Label the x-axis as "`Survived` (0 = No, 1 = Yes)" and the y-axis as "Count".

Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the `Fare` and `Age` of passengers.
- Use orange for the data points.
- Title the plot as "Fare vs Age".
- Label the x-axis as "`Age`" and the y-axis as "`Fare`".

Note: Refer to the displayed plot in the sample test cases for better understanding.

Sample Test Cases

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```

```
## Set up the figure for 5 subplots
# fig, axes = plt.subplots(3, 2,
figsize=(12, 12))

import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the
CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2,
figsize=(12, 12))

import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the
CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2,
figsize=(12, 12))

# write the code..
#plot 1: Count of passenger by class
axes[0,0].bar(df['Pclass'].value_coun
ts().index,
df['Pclass'].value_counts(),color='sk
yblue')
axes[0,0].set_title("Passenger Class
Distribution")
axes[0,0].set_xlabel("Pclass")
axes[0,0].set_ylabel("Count")

#plot 2: Gender Distribution
axes[0,1].pie(df['Gender'].value_coun
ts(),labels=df['Gender'].value_counts
().index,autopct='%1.1f%%',colors=
['lightblue','lightcoral'])
axes[0,1].set_title("Gender
Distribution")

#plot 3: Age Distribution
axes[1,0].hist(df['Age'].dropna(),bin
s=8,color='lightgreen',edgecolor='bla
ck')
axes[1,0].set_title("Age
Distribution")
axes[1,0].set_xlabel("Age")
axes[1,0].set_ylabel("Frequency")

#plot 4: Survival Count
axes[1,1].bar(df['Survived'].value_co
unts().index,df['Survived'].value_cou
nts(),color=
['lightblue','lightcoral'])
axes[1,1].set_title("Survival Count")
axes[1,1].set_xlabel("Survived (0 =
No, 1 = Yes)")
axes[1,1].set_ylabel("Count")

#plot 5: Fare vs Age
axes[2,0].scatter(df['Age'],df['Fare'
],color='orange',edgecolors='black')
axes[2,0].set_title("Fare vs Age")
axes[2,0].set_xlabel("Age")
axes[2,0].set_ylabel("Fare")

plt.tight_layout()
plt.show()
```

5.2.2. Histogram of passenger information o...

03:59



Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases



Histogram...



Submit

Explorer

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
(), inplace=True)
9 data['Embarked'].fillna(data['Embarke
d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
inplace=True)
11
12 # Convert categorical features to
numeric
13 data['Sex'] =
data['Sex'].map({'male': 0,
'female': 1})
14 data = pd.get_dummies(data, columns=
['Embarked'], drop_first=True)
15 plt.hist(data["Age"].dropna(),
bins=30, edgecolor= "k")
16 plt.xlabel("Age")
17 plt.ylabel("Frequency")
18 plt.title("Age Distribution")
19 plt.show()
```

Debugger

5.2.3. Bar plot of survival rate of passengers

01:29



Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title "Survival Count" to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases



Explorer

BarPlotOfs...



Submit

Debugger

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
(), inplace=True)
9 data['Embarked'].fillna(data['Embarke
d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
inplace=True)
11
12 # Convert categorical features to
numeric
13 data['Sex'] =
data['Sex'].map({'male': 0,
'female': 1})
14 data = pd.get_dummies(data, columns=
['Embarked'], drop_first=True)
15 survival_counts =
data['Survived'].value_counts().sort_
index()
16 survival_counts.plot(kind='bar')
17 plt.title("Survival Count")
18 plt.xlabel("Survived")
19 plt.ylabel("Count")
20 plt.show()
```

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "Survival by Gender" to the chart.
4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

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```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
   Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
   (), inplace=True)
9 data['Embarked'].fillna(data['Embarke
   d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
   inplace=True)
11
12 # Convert categorical features to
   numeric
13 data['Sex'] =
   data['Sex'].map({'male': 0,
   'female': 1})
14 data = pd.get_dummies(data, columns=
   ['Embarked'], drop_first=True)
15 survival_gender =
   data.groupby(['Sex',
   'Survived']).size().unstack()
16 ax =
   survival_gender.plot(kind='bar',
   stacked=True)
17 plt.title("Survival by Gender")
18 plt.xlabel("Gender")
19 plt.ylabel("Count")
20 plt.legend(["Not Survived",
   "Survived"])
21 plt.show()

```


5.2.5. Bar Plot for Survival by Pclass

01:42

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "Survival by Pclass" to the chart.
4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,

```

Note:

- Refer to the visible test case for better reference.
- Ensure you use the **groupby()** function with **value_counts()** to count the survivors and non-survivors for each **Pclass**.
- Do not manually use **size()** or **unstack()** without **value_counts()**. Use the **value_counts()** method for counting survival status directly.

Sample Test Cases

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BarPlotOfS...



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Explorer

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
6 Dataset.csv')
7
8 # Data Cleaning
9 data['Age'].fillna(data['Age'].median
10 (), inplace=True)
11 data['Embarked'].fillna(data['Embarke
12 d'].mode()[0], inplace=True)
13
14 # Convert categorical features to
15 numeric
16 data['Sex'] =
17 data['Sex'].map({'male': 0,
18 'female': 1})
19 data = pd.get_dummies(data, columns=
20 ['Embarked'], drop_first=True)
21 survival_pclass =
22 data.groupby(['Pclass',
23 'Survived']).size().unstack()
24 survival_pclass.plot(kind='bar',
25 stacked=True)
26 plt.title("Survival by Pclass")
27 plt.xlabel("Pclass")
28 plt.ylabel("Count")
29 plt.legend(["Not Survived",
30 "Survived"])
31 plt.show()

```

Debugger

5.2.6. Bar Plot for Survival by Embarked

02:21

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked" to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

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Explorer

BarPlotOfS...

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Submit

Debugger

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
   Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
   (), inplace=True)
9 data['Embarked'].fillna(data['Embarke
   d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
   inplace=True)
11
12 # Convert categorical features to
   numeric
13 data['Sex'] =
   data['Sex'].map({'male': 0,
   'female': 1})
14 data = pd.get_dummies(data, columns=
   ['Embarked'], drop_first=True)
15 survival_embarked =
   data.groupby(['Embarked_Q',
   'Survived']).size().unstack()
16 survival_embarked.plot(kind='bar',
   stacked=True)
17 plt.title("Survival by Embarked")
18 plt.xlabel("Embarked")
19 plt.ylabel("Count")
20 plt.legend(["Not Survived",
   "Survived"])
21 plt.show()

```


Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to "Age by Pclass".
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as 'Pclass' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Tic
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thaye
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 31
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fem
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.862
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

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Explorer

BoxPlotFor...

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Submit

Debugger

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
   Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
   (), inplace=True)
9 data['Embarked'].fillna(data['Embarke
   d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
   inplace=True)
11
12 # Convert categorical features to
   numeric
13 data['Sex'] =
   data['Sex'].map({'male': 0,
   'female': 1})
14 data = pd.get_dummies(data, columns=
   ['Embarked'], drop_first=True)
15 data.boxplot(column='Age',
   by='Pclass')
16 plt.title("Age by Pclass")
17 plt.suptitle('')
18 plt.xlabel("Pclass")
19 plt.ylabel("Age")
20 plt.show()

```

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to "Age by Survival".
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as 'Survived' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,34,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

+

Explorer

BoxPlotFor...

▶

Submit

Debugger

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
   Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
   (), inplace=True)
9 data['Embarked'].fillna(data['Embarke
   d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
   inplace=True)
11
12 # Convert categorical features to
   numeric
13 data['Sex'] =
   data['Sex'].map({'male': 0,
   'female': 1})
14 data = pd.get_dummies(data, columns=
   ['Embarked'], drop_first=True)
15 data.boxplot(column='Age',
   by='Survived')
16 # Customize titles and labels
17 plt.title("Age by Survival")
18 plt.suptitle('')
19 plt.xlabel("Survived")
20 plt.ylabel("Age")
21 # Display the plot
22 plt.show()
```


Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

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Explorer

BoxPlotFor...

⏮

Submit

Debugger

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
(), inplace=True)
9 data['Embarked'].fillna(data['Embarke
d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
inplace=True)
11
12 # Convert categorical features to
numeric
13 data['Sex'] =
data['Sex'].map({'male': 0,
'female': 1})
14 data = pd.get_dummies(data, columns=
['Embarked'], drop_first=True)
15 data.boxplot(column='Fare',
by='Pclass')
16 plt.title("Fare by Pclass")
17 plt.suptitle('') # Removes the
default "Boxplot grouped by Pclass"
subtitle
18 plt.xlabel("Pclass")
19 plt.ylabel("Fare")
20 # Display the plot
21 plt.show()
```

5.2.10. Scatter Plot for Age vs. Fare

02:44

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to "Age vs. Fare".
3. Label the x-axis as 'Age' and the y-axis as 'Fare'.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

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Explorer

AgeFareSc...

▶

Submit

Debugger

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
   Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
   (), inplace=True)
9 data['Embarked'].fillna(data['Embarke
   d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
   inplace=True)
11
12 # Convert categorical features to
   numeric
13 data['Sex'] =
   data['Sex'].map({'male': 0,
   'female': 1})
14 data = pd.get_dummies(data, columns=
   ['Embarked'], drop_first=True)
15 plt.scatter(data['Age'],
   data['Fare'])
16 plt.title("Age vs. Fare")
17 plt.xlabel("Age")
18 plt.ylabel("Fare")
19 # Display the plot
20 plt.show()

```


5.2.11. Scatter Plot for Age vs. Fare by Surviv... 01:24

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

```

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ti
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thay
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",fe
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.86
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,34990
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

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AgeFareSc...



Submit

Explorer



```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-
Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median
(), inplace=True)
9 data['Embarked'].fillna(data['Embarke
d'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1,
inplace=True)
11
12 # Convert categorical features to
numeric
13 data['Sex'] =
data['Sex'].map({'male': 0,
'female': 1})
14 data = pd.get_dummies(data, columns=
['Embarked'], drop_first=True)
15 colors = data['Survived'].map({0:
'red', 1: 'blue'})
16 # Create the scatter plot for Age vs
Fare using the c argument for colors
17 plt.scatter(data['Age'],
data['Fare'], c=colors)
18 # Add title and labels
19 plt.title("Age vs. Fare by Survival")
20 plt.xlabel("Age")
21 plt.ylabel("Fare")
22 # Display the plot
23 plt.show()

```

Debugger

